

Introduction to Stochastic Calculus

Derek Chen

Abstract

This paper explores the distinctions between ordinary calculus and stochastic calculus, focusing on Brownian motion, Itô’s lemma, and applications in financial modelling.

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1 Introduction

In this essay, I explore the stochastic models underpinning a Monte Carlo simulation that I recently constructed, focusing on the distinctions between ordinary and stochastic calculus. As part of my supercurricular studies in Year 12, I became intrigued by the probabilistic behaviour predicted by the model and began studying stochastic calculus to better understand how uncertainty influences mathematical systems.

This paper focuses on intuitive explanations of key concepts such as random walk theory, Brownian motion, Itô's lemma, and stochastic differential equations (SDEs), rather than formal proofs.

2 Ordinary Calculus

2.1 Ordinary Differentiation

Before exploring stochastic calculus, we must understand ordinary calculus. Using the definition of the derivative from first principles,

$$F'(x) = \lim_{dx \rightarrow 0} \frac{F(x + dx) - F(x)}{dx},$$

we can find the derivative of any function. Typically, all second-order differential terms such as dx^2 vanish in the limit. This follows from the Taylor series expansion

$$F(x + dx) = F(x) + F'(x) dx + \frac{1}{2}F''(x)(dx)^2 + \dots,$$

so that

$$F'(x) = \lim_{dx \rightarrow 0} \frac{F(x) + F'(x) dx + \frac{1}{2}F''(x)(dx)^2 + \dots - F(x)}{dx}.$$

Hence, it is concluded that the second-order differential terms vanish in ordinary calculus. As we will see, this property does *not* carry over to stochastic calculus.

2.2 Ordinary Integration

Riemann integrals define the area under a curve and, through the Fundamental Theorem of Calculus, are closely related to antiderivatives. However, not all integrals can be analytically solved, hence numerical integration was developed to compute the sum of rectangles.

$$\int_a^b F(x) dx \approx \sum_{i=0}^{n-1} F(x_i) \Delta x, \quad \Delta x = \frac{b-a}{n}.$$

Note that this uses the left-endpoint for each rectangle.

3 Stochastic Models

3.1 Overview

Ordinary calculus is highly effective for deterministic systems. However, when randomness is involved in a system, deterministic methods can no longer capture future

behaviour adequately. This is because random variables dictate the outcomes of such models and throughout this essay, I will refer to a stochastic model called Brownian motion. One of the clearest applications I have encountered in stochastic calculus is predicting future stock prices, and I will use that reference throughout.

3.2 Brownian Motion

Consider a coin flipped repeatedly, where the outcomes of heads or tails occur with equal probability. Each head gains £1; each tail loses £1. Let X_i be the outcome on the i -th toss and define the partial sum $S_n = \sum_{i=1}^n X_i$, so that $E[S_n] = 0$ and $\text{Var}[S_n] = n$. This is a *random walk*: each increment is independent of the others, forming a stochastic process (Benjamin, 2025). The process S_n satisfies the *Markov property*: the distribution of the next state depends only on the current position; it has no memory beyond where it is currently.

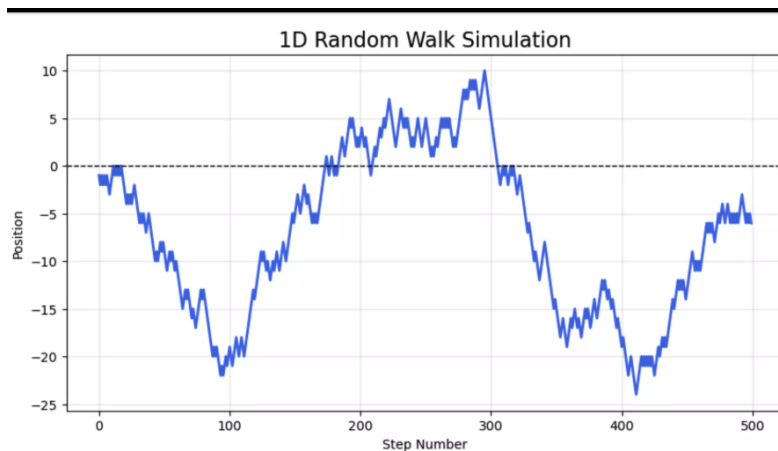


Figure 1: Simulated random walk

Brownian motion is the limit of a rescaled symmetric random walk. It was first observed by Robert Brown in 1827 when he examined how random molecular collisions caused pollen particles to move in a continuous, unpredictable manner (**Walia'2021'brownian**). The French mathematician Louis Bachelier later linked this phenomenon to financial markets, observing that price fluctuations over small time intervals are independent of the current price and of historical price movements (Index Fund Advisors, 2025). Combining these assumptions with the Central Limit Theorem, Bachelier argued that price behaviour could be modelled by a normal distribution (Gowers, 2009).

A standard Brownian motion W_t satisfies the following properties (Lalley, n.d.):

1. $W_0 = 0$
2. W_t is a continuous function, yet it is non-differentiable for all values
3. W_t has independent increments: for all $t_1 < t_2$, $W_{t_2} - W_{t_1} \sim \mathcal{N}(0, t_2 - t_1)$
4. W_t is a random variable

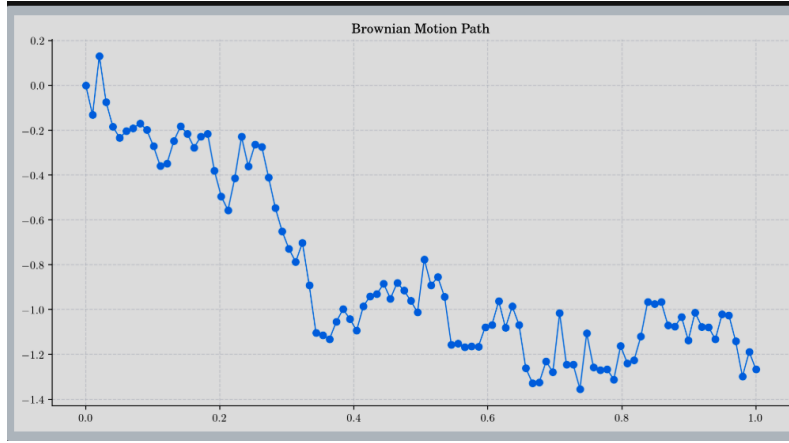


Figure 2: Sample paths of Brownian motion

3.3 Itô's Lemma

Itô's lemma is an essential tool in stochastic calculus for determining how a stochastic process evolves over time (Gheorghiu, 2011). Often regarded as the *stochastic chain rule*, it plays a central role in solving SDEs.

As noted above, not all second-order differentials vanish when a random process like Brownian motion is involved. As $dt \rightarrow 0$, the terms $dt^2 \rightarrow 0$ and $dt dW_t \rightarrow 0$. Surprisingly, however, $dW_t^2 = dt$. This happens as a result of the non-zero *quadratic variation* of Brownian motion. Quadratic variation measures the irregularity of a path. Formally, the quadratic variation of Brownian motion over the partition $0 = t_0 < t_1 < \dots < t_n = t$, is the limit of the sum of squared increments (Gheorghiu, 2011):

$$[W]_t = \lim_{\|\Pi\| \rightarrow 0} \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2 = t.$$

Because $\Delta W_t \sim \mathcal{N}(0, \Delta t)$, the increment ΔW_t is of order $\sqrt{\Delta t}$ and cannot be neglected, giving $dW_t^2 = dt$.

We now derive Itô's lemma. For a time-dependent function of Brownian motion $F(t, W_t)$, the Taylor series expansion (Levas, 2016) gives

$$\begin{aligned} F(t + dt, W_t + dW_t) &= F(t, W_t) + \frac{\partial F}{\partial t} dt + \frac{\partial F}{\partial W_t} dW_t \\ &\quad + \frac{1}{2} \frac{\partial^2 F}{\partial t^2} (dt)^2 + \frac{\partial^2 F}{\partial t \partial W_t} dt dW_t + \frac{1}{2} \frac{\partial^2 F}{\partial W_t^2} (dW_t)^2 + \dots \end{aligned}$$

Applying the identities established previously, and discarding all higher-order terms, we obtain **Itô's lemma**:

$$dF = \left(\frac{\partial F}{\partial t} + \frac{1}{2} \frac{\partial^2 F}{\partial W_t^2} \right) dt + \frac{\partial F}{\partial W_t} dW_t.$$

3.4 Solving SDEs

Stochastic calculus does not provide a single deterministic outcome. for example, it will not directly provide a price of a future stock. Instead, it helps demonstrate how a stock portfolio *may* evolve and, through repeated simulation, helps estimate the long-run probability distribution of future prices (Certificate in Quantitative Finance, 2025).

Consider the SDE defined by

$$F(t, W_t) = 2e^t + W_t^2.$$

Here, W_t^2 denotes the square of the Brownian motion process at time t . By computing the required partial derivatives,

$$\frac{\partial F}{\partial t} = 2e^t, \quad \frac{\partial F}{\partial W_t} = 2W_t, \quad \frac{\partial^2 F}{\partial W_t^2} = 2,$$

and substituting into Itô's lemma, we obtain

$$dF(t, W_t) = (2e^t + 1) dt + 2W_t dW_t.$$

The coefficient of dt is the *drift term*: it captures the average, deterministic trend of the portfolio price over time. The coefficient of dW_t is the *diffusion term*: it represents the volatility driven by random Brownian fluctuations. Together, these terms show that no matter how small the time step, randomness and uncertainty are always embedded in the model.

4 Conclusion

The real world is far more complex and dynamic than deterministic models suggest, so modelling uncertainty is an effective way to draw reasonable conclusions. SDEs combine ordinary differential equations with random noise, allowing mathematicians to model systems governed by both predictable trends and unpredictable fluctuations (Certificate in Quantitative Finance, 2025). This is particularly valuable when modelling future stock prices, since many SDE-based models naturally produce processes with the Markov property.

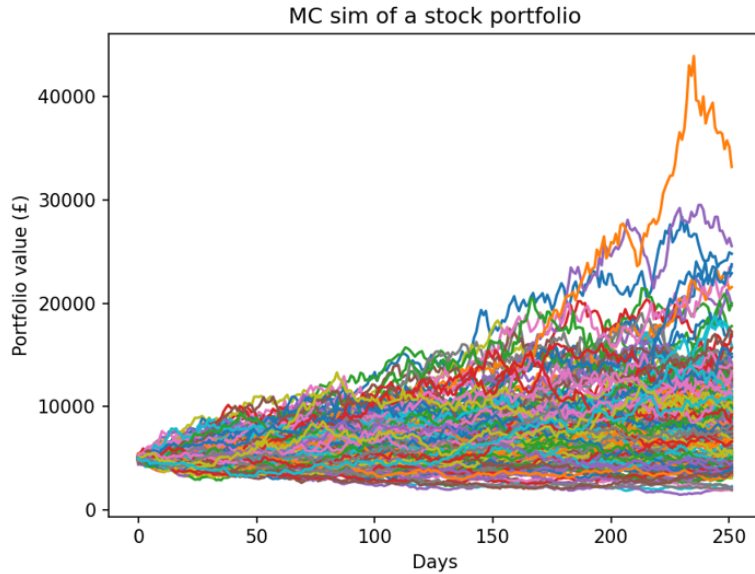


Figure 3: Monte Carlo simulation of stock portfolio price paths

A common application of SDEs is in the Monte Carlo method (Figure 3), which simulates complex financial instruments subject to multiple sources of uncertainty.

Each simulation path is generated by an SDE, and by running many such paths one can approximate the full distribution of future outcomes and estimate expected values. The Central Limit Theorem guarantees that the sample mean converges to a normal distribution as the number of simulations grows (Walia, 2021).

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